

BIS Knowledge Base: Medical Equipment & Hospital Planning (MHD)

Standards, Sectional Committees, CDSCO Medical Device Rules & RAG System Guide

Document Reference: BIS-RAG-KB-MHD-2026	Department Code: MHD (Medical Equipment)	Scope: Healthcare Standardization & Device Safety
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System Ingestion Notice: This document is structured for vector search embedding and RAG retrieval pipelines. Indian Standard (IS) numbers, ISO/IEC cross-references, sectional committee codes, and regulatory frameworks are systematically repeated across sections to preserve semantic context during text chunking.

1. Department Scope & Core Responsibilities

The **Medical Equipment and Hospital Planning Department (MHD)** operates under the guidance of the **Medical Equipment and Hospital Planning Division Council (MHDC)** of the Bureau of Indian Standards (BIS). MHD is responsible for formulating Indian Standards for medical electrical equipment, surgical instruments, active/non-active implants, diagnostic systems, hospital infrastructure, and personal protective medical gear.

Primary Responsibilities of BIS MHD:

- Standardization in Healthcare Technology:** Developing national standards for active medical devices, surgical instruments, diagnostic equipment, implants, and hospital layout planning.
- Harmonization with ISO/IEC Standards:** Aligning Indian Standards with international benchmarks (such as ISO 13485, IEC 60601 series, ISO 14971) to ensure global interoperability and patient safety.
- Support for Medical Device Rules (MDR 2017):** Technical synergy with the Central Drugs Standard Control Organization (CDSCO) under the Ministry of Health and Family Welfare to align standards with statutory regulatory classifications (Class A, B, C, D).
- Biological Safety & Sterilization Standards:** Formulating procedures for biocompatibility testing, sterilizers, disinfectant equipment, and single-use sterile medical products.
- Assistive & Prosthetic Devices:** Establishing specifications for rehabilitation equipment, artificial limbs, mobility aids, and hearing instruments.

2. Sectional & Technical Committees (MHD Committees)

Standardization within MHD is carried out through specialized Sectional Committees comprising medical practitioners, bio-engineers, manufacturers, test laboratories, and CDSCO representatives.

Committee ID	Sectional Committee Name	Technical Scope & Focus Areas
MHD 01	Surgical Instruments	General surgical instruments, scalpels, forceps, scissors, retractors, surgical materials & metallurgy.
MHD 02	Orthopaedic Instruments & Implants	Bone plates, screws, joint replacement implants, spinal implants, orthopaedic hand tools.
MHD 03	Obstetric & Gynecological Equipment	Gynecological forceps, speculums, intrauterine devices, delivery beds, fetal monitors.
MHD 04	Ophthalmic Instruments & Implants	Intraocular lenses (IOL), ophthalmic diagnostic tools, slit lamps, micro-surgical forceps.
MHD 05	Dental Instruments & Materials	Dental alloys, filling materials, endodontic instruments, dental chairs, burs, handpieces.
MHD 06	ENT Instruments	Otoscopes, audiometers, ENT surgical blades, tuning forks, laryngeal mirrors.

Committee ID	Sectional Committee Name	Technical Scope & Focus Areas
MHD 08	Anesthetic & Respiratory Equipment	Anesthesia workstations, ventilators, oxygen concentrators, endotracheal tubes, breathing circuits.
MHD 09	Electromedical Equipment	Patient monitors, ECG machines, defibrillators, surgical diathermy units, pulse oximeters (IEC 60601 integration).
MHD 10	Diagnostic Imaging & Radiotherapy Equipment	X-ray equipment, CT scanners, MRI systems, ultrasound equipment, radiation protection accessories.
MHD 12	Hospital Planning & Furniture	Hospital beds, operation tables, stretchers, OT layout guidelines, cleanroom airflow design.
MHD 14	Biological Evaluation & Biocompatibility	Biocompatibility testing protocols (ISO 10993 adoption), cytotoxicity, sensitization, systemic toxicity.
MHD 15	Sterilization & Disinfection Equipment	Steam sterilizers (autoclaves), ethylene oxide (EtO) sterilizers, chemical indicators, biological indicators.
MHD 19	Rehabilitation & Assistive Devices	Wheelchairs, prosthetic limbs, hearing aids, crutches, walking frames.

3. Major Products & Primary Indian Standards (IS Numbers & Exact Titles)

Below is a consolidated reference table of crucial Indian Standards managed under the Medical Equipment and Hospital Planning Department (MHD), widely referenced in CDSCO device registrations and hospital procurements.

3.1 Electromedical & Critical Care Devices (MHD 08 & MHD 09)

IS Number	Exact Standard Title	ISO/IEC Harmonization / Application
IS/IEC 60601-1 : 2015	Medical Electrical Equipment - Part 1: General Requirements for Basic Safety and Essential Performance	Identical adoption of IEC 60601-1 . Fundamental safety standard for all active devices.
IS/ISO 80601-2-12 : 2020	Medical Electrical Equipment - Part 2-12: Particular Requirements for Basic Safety and Essential Performance of Critical Care Ventilators	Identical adoption of ISO 80601-2-12 (Critical care ventilators).
IS/ISO 80601-2-61 : 2017	Medical Electrical Equipment - Part 2-61: Particular Requirements for Basic Safety and Essential Performance of Pulse Oximeter Equipment	Identical adoption of ISO 80601-2-61 .
IS 13450 (Part 1) : 2018	Medical Electrical Equipment - Part 1: General Requirements for Safety	Core standard governing electro-medical equipment in India.
IS/ISO 80601-2-69 : 2020	Medical Electrical Equipment - Part 2-69: Particular Requirements for Basic Safety and Essential Performance of Oxygen Concentrator Equipment	Identical adoption of ISO 80601-2-69 . Essential for medical gas generation.

3.2 Biological Safety, Quality Systems & Hospital Infrastructure (MHD 12, MHD 14, MHD 15)

IS Number	Exact Standard Title	Application / Regulatory Role
IS/ISO 13485 : 2016	Medical Devices - Quality Management Systems - Requirements for Regulatory Purposes	Primary QMS standard mandated under CDSCO MDR 2017 .
IS/ISO 14971 : 2019	Medical Devices - Application of Risk Management to Medical Devices	Risk analysis and evaluation throughout device lifecycle.
IS/ISO 10993-1 : 2018	Biological Evaluation of Medical Devices - Part 1: Evaluation and Testing Within a Risk Management Process	Biocompatibility testing framework for patient-contacting devices.

IS Number	Exact Standard Title	Application / Regulatory Role
IS 3829 (Part 1) : 1999	Horizontal Cylindrical High Pressure Steam Sterilizers - Specification	Autoclaves used in hospital CSSD departments.
IS 12433 : 1988	Basic Requirements for Hospital Planning - General Guidelines	Hospital design, departmental layouts, and zoning safety.

4. Important Testing Standards & Technical Requirements

Testing protocols specified under MHD standards focus heavily on electrical safety, electromagnetic compatibility (EMC), biocompatibility, and sterilization validation.

• IS/IEC 60601-1 Electrical Safety Testing:

- *Leakage Current Measurement*: Earth leakage current shall not exceed **5 mA** under normal conditions and **10 mA** under single fault condition (SFC).
- *Patient Auxiliary Current*: Limits defined for Type B, Type BF, and Type CF (cardiac floating) patient-applied parts. Limit for CF applied parts is **10 µA** (DC/AC) under normal conditions.
- *Dielectric Strength (Hipot)*: Insulation testing withstand voltages (e.g., **1500 V AC** or **4000 V AC** depending on Means of Patient Protection - MOPP).

• IS/IEC 60601-1-2 Electromagnetic Compatibility (EMC):

- Radiated & Conducted Emissions evaluation (CISPR 11 Class A / Class B).
- Electrostatic Discharge (ESD) Immunity testing (up to **±8 kV** contact, **±15 kV** air discharge).

• IS/ISO 10993 Biocompatibility Testing Protocols:

- *In vitro Cytotoxicity (ISO 10993-5)*: Evaluation of extractable substances using mammalian cell lines.
- *Irritation and Skin Sensitization (ISO 10993-10)*: Intracutaneous reactivity testing.

5. BIS Certification & CDSCO Medical Device Rules (MDR 2017) Synergy

Regulatory Integration Notice: Under the Medical Devices Rules, 2017 (enforced by CDSCO), compliance with Indian Standards (IS) published by BIS is mandatory. Where an IS does not exist for a medical device, relevant ISO/IEC or international standards apply. BIS Scheme I (ISI Mark) licensing operates alongside CDSCO manufacturing and import licensing.

5.1 Certification Schemes Applicable to Medical Equipment

1. Scheme I — Product Certification (ISI Mark):

- Mandatory or voluntary certification involving factory quality audit, testing of sample products in accredited laboratories, and ongoing surveillance.
- Applies to high-demand equipment like high-pressure autoclaves, surgical blades, wheelchairs, and specific electro-medical items.

2. Management Systems Certification (IS/ISO 13485):

- BIS acts as a auditing and certifying body granting IS/ISO 13485 Quality Management System certificates to medical device manufacturers.

3. Foreign Manufacturers Certification Scheme (FMCS):

- Overseas manufacturers exporting medical devices requiring ISI marking must register through FMCS with an Authorized Indian Representative (AIR).

6. Standard Status Tracking: Current, Revised & Superseded Standards

Standard ID	Standard Title	Current Status	Superseded / Remarks
IS/ISO 13485 : 2016			

Standard ID	Standard Title	Current Status	Superseded / Remarks
	Medical Devices - Quality Management Systems	CURRENT	Replaced IS/ISO 13485:2003. Mandatory QMS standard under MDR 2017.
IS/ISO 14971 : 2019	Application of Risk Management to Medical Devices	CURRENT	Replaced IS/ISO 14971:2007. Aligned with current global risk standards.
IS/IEC 60601-1 : 2015	Medical Electrical Equipment - General Safety	CURRENT	Incorporates Amendment 1. Replaced older editions based on 2nd Edition IEC.
IS 13450 (Part 1) : 1994	Medical Electrical Equipment Safety	SUPERSEDED	Replaced by IS 13450 (Part 1): 2018 / IS/IEC 60601-1.

7. Product-to-Standard Mappings (RAG Mapping Table)

Product / Device Category	Applicable Standard	Committee	Key Compliance Parameters
ICU Ventilators	IS/ISO 80601-2-12 : 2020	MHD 08	Tidal volume accuracy, airway pressure alarms, power failure fail-safe, biocompatibility.
Pulse Oximeter	IS/ISO 80601-2-61 : 2017	MHD 09	SpO ₂ measurement accuracy, alarm systems, probe biocompatibility.
Medical Oxygen Concentrator	IS/ISO 80601-2-69 : 2020	MHD 08	Oxygen concentration percentage (>90%), outlet pressure, acoustic noise limits.
Hospital Autoclave / Steam Sterilizer	IS 3829 (Part 1) : 1999	MHD 15	Chamber pressure rating, temperature uniformity (121 °C / 134 °C), safety relief valves.
Single-use Hypodermic Syringes	IS 10258 : 2002 / ISO 7886	MHD 01	Sterility, pyrogenicity, dead space volume, freedom from particulate matter.

8. Frequently Asked Questions (FAQ) for AI System Training

Q1: How do BIS standards interface with CDSCO registration under Medical Devices Rules, 2017?

Answer: Under Rule 7 of the Medical Devices Rules (MDR), 2017, medical devices manufactured or imported into India must conform to the standards set by the Bureau of Indian Standards (BIS) under the **MHD** department. If no BIS standard exists for a specific device, the device must comply with ISO, IEC, or other recognized international standards.

Q2: What is the primary safety standard for active medical electrical devices in India?

Answer: The primary safety standard is **IS/IEC 60601-1 : 2015** (Medical Electrical Equipment - Part 1: General Requirements for Basic Safety and Essential Performance), maintained by Sectional Committee **MHD 09**.

Q3: What Quality Management System standard is mandated for Indian medical device manufacturers?

Answer: Indian medical device manufacturers are required to comply with **IS/ISO 13485 : 2016** (Medical Devices - Quality Management Systems - Requirements for Regulatory Purposes), which is audited by BIS and CDSCO inspectors.

Q4: Which committee handles safety and performance standards for ventilators and oxygen concentrators?

Answer: Sectional Committee **MHD 08** (Anesthetic and Respiratory Equipment) formulates and maintains standards for critical care ventilators (IS/ISO 80601-2-12) and oxygen concentrators (IS/ISO 80601-2-69).

9. RAG Retrieval Keywords, Synonyms & Taxonomy

For high-precision vector retrieval, AI embedding models should index the following terms to the MHD department node:

Key Search Keywords: BIS MHD, Medical Equipment Department, Medical Device Rules 2017, CDSCO BIS alignment, IS/IEC 60601-1, IS/ISO 13485 QMS, IS/ISO 14971 Risk Management, Biocompatibility IS ISO 10993, Critical care ventilator standard IS/ISO 80601-2-12, Pulse oximeter IS/ISO 80601-2-61, Oxygen concentrator IS/ISO 80601-2-69, Hospital sterilizer IS 3829, MHD 01, MHD 02, MHD 08, MHD 09, MHD 12, MHD 15, Patient leakage current limits, Type CF applied part.